Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 19: At Most One, At Least One

One-to-One and Onto Functions

Chartrand & Zhang, Section 5.4

Name: _____ Date: _____

Where we are

A function sends each element of A somewhere. Today we ask two questions about where things land: does anything get hit *twice*, and does everything get hit *at all*?

By the end of today you should be able to

1. Prove a function is one-to-one using the contrapositive form.
2. Prove a function is onto by producing a preimage.
3. Disprove either with a specific witness.

1. One-to-one

Definition 5.45

$f : A \to B$ is **one-to-one** (or **injective**) if every two distinct elements of A have distinct images: if $a \neq b$ then $f(a) \neq f(b)$.

That form is awkward to work with — it starts from an inequality. So we use its contrapositive:

The working form

f is one-to-one if whenever $f(a) = f(b)$, it follows that $a = b$.

Theorem 1.46, five weeks later

These two are logically equivalent because an implication and its contrapositive always are. In Week 2 that was a truth-table exercise. Here it is the difference between a proof you can write and one you cannot: starting from $f(a) = f(b)$ gives you an *equation* to manipulate, while starting from $a \neq b$ gives you almost nothing.

Worked (Result 5.49). Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 5x - 3$.

Proof. Assume $f(a) = f(b)$. Then $5a - 3 = 5b - 3$. Adding 3 gives $5a = 5b$, and dividing by 5 gives $a = b$. Hence f is one-to-one. ■

To disprove: exhibit two distinct elements with the same image.

Function	One-to-one?	Witness
$f(x) = x^2 + 1$ on $\mathbb{R}$	No	$f(1) = f(-1) = 2$
$h(x) = x^2 - 3x + 1$ on $\mathbb{R}$	No	$h(0) = h(3) = 1$
$f(n) = 2n$ on $\mathbb{Z}$	Yes	—

Where the second witness came from

$h(x) = x^2 - 3x + 1 = x(x-3) + 1$. The product $x(x-3)$ is zero at $x = 0$ and $x = 3$, so both give $h = 1$.
Rewriting the formula produced the counterexample. Hunting blindly rarely does.

2. A counting consequence

Result (5.3)

If $f : A \to B$ is one-to-one and A, B are finite, then $|B| \geq |A|$.

Distinct elements need distinct images, so $|A|$ inputs require at least $|A|$ available outputs.

The contrapositive says: if $|B| < |A|$, there is **no** one-to-one function from A to B.

That is the pigeonhole principle

Put $|A|$ items into $|B|$ boxes with $|B| < |A|$, and some box gets two items. You will meet it by name in Chapter 8, where it does surprisingly heavy work.
For infinite sets it fails. $f(n) = 2n$ is a one-to-one function from $\mathbb{Z}$ onto a *proper subset* of $\mathbb{Z}$ (Example 5.48). Infinite sets can inject into parts of themselves — which is the beginning of the theory of cardinality.

3. Onto

Definition — onto

$f : A \to B$ is **onto** (or **surjective**) if every element of B is the image of at least one element of A — equivalently, if $f(A) = B$, that is, the range equals the codomain.

How to prove it

Let b be an *arbitrary* element of the codomain. **Produce** an a in the domain with $f(a) = b$, and verify it.

Finding that a is usually done by solving $f(x) = b$ for x — as scratch work, before writing the proof.

Worked. $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 5x - 3$, is onto.

Scratch: solve $5x - 3 = b$ to get $x = (b+3)/5$.

Proof. Let $b \in \mathbb{R}$. Set $a = (b+3)/5$, which is a real number. Then $f(a) = 5\left(\frac{b+3}{5}\right) - 3 = b$. So f is onto. ■

To disprove: name an element of the codomain that is the image of nothing.

Function	Onto?	Witness
$f(x) = x^2 + 1$, $\mathbb{R} \to \mathbb{R}$	No	0 is not an image; $x^2 + 1 \geq 1$.
$f(n) = 2n$, $\mathbb{Z} \to \mathbb{Z}$	No	1 is not an image — no odd integer is.
$f(x) = x^3$, $\mathbb{R} \to \mathbb{R}$	Yes	Given b, take $a = \sqrt[3]{b}$.

4. The codomain decides

Consider $f(x) = \sqrt{x}$ three times, with the rule unchanged.

Function	Onto?	Why
$f : \mathbb{R}^+ \to \mathbb{R}^+$	Yes	Given $r > 0$, $f(r^2) = r$.
$g : \mathbb{R}^+ \to \mathbb{R}$	No	$\sqrt{x} > 0$, so no $y \leq 0$ is an image.

Same formula, different function

f and g have the same rule and the same domain. Only the **codomain** changed, and that changed the answer.
This is why Monday's warning matters: the codomain is *declared*, not computed. Asking "is this function onto?" is meaningless until someone tells you the codomain.

Exit ticket

1. Prove $f : \mathbb{R} \to \mathbb{R}$, $f(x) = 3x + 7$, is one-to-one.

2. Give a witness showing $f(n) = n^2$ on $\mathbb{Z}$ is not one-to-one.

Before Friday

Read Section 5.4, pp. 174–178 — bijective functions and inverses.

Self-check (not collected). Section 5.4, Exercises 1, 3, and 9 — odd-numbered, answers in the back.

Homework 7 is due Friday, October 2, 11:59 p.m.

Questions I still have